Use all the lifting eyes when lifting OPU.

C2.8 Industrial Engine

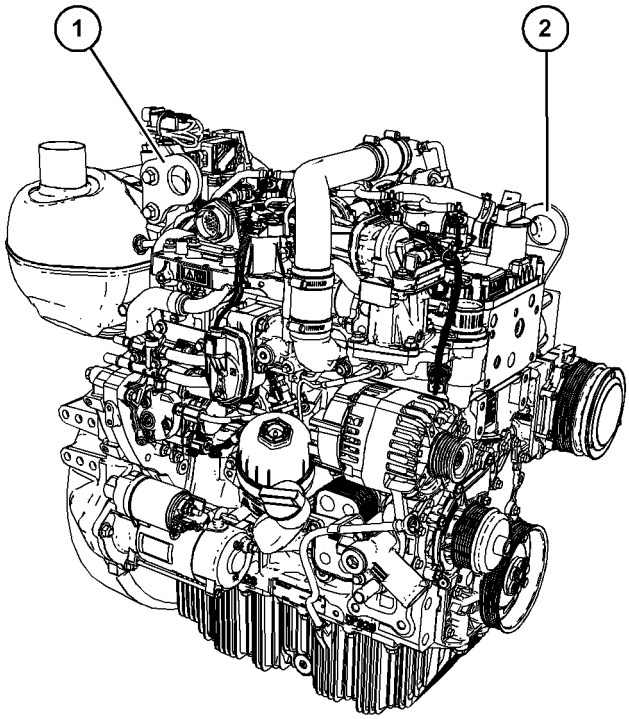


Illustration 48 g06458398

Typical example

(1) Rear lifting eye
(2) Front lifting eye

Use both lifting eyes when lifting engine assembly.

Caterpillar recommends that model-specific cradles are used to store and transport engines. Supporting engines on the sump/oil pan is not recommended under any circumstances. This action is for both safety and quality reasons.

C2.8 Open Power Unit (OPU)

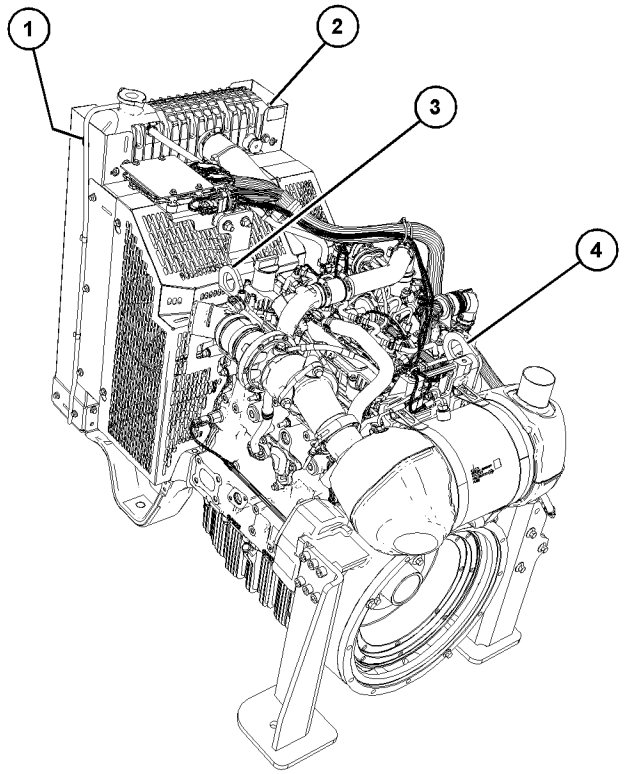


Illustration 49 g06742366

Typical example

(1) Radiator lifting eye (1 of 2)
(2) Radiator lifting eye (2 of 2)
(3) Front lifting eye
(4) Rear lifting eye

Use both lifting eyes when lifting engine assembly.

Use both lifting eyes when lifting radiator assembly.

Use all the lifting eyes when lifting the OPU.

i07676027

Product Storage

SMCS Code: 7002

If the engine will not be started for several weeks, the lubricating oil will drain from the cylinder walls and from the piston rings. Rust can form on the cylinder liner surface. Rust on the cylinder liner surface will cause increased engine wear and a reduction in engine service life.

To help prevent excessive engine wear, use the following guidelines:

- Complete all the lubrication recommendations that are listed in the Operation and Maintenance Manual, “Maintenance Interval Schedule”.
- If freezing temperatures are expected, check the cooling system for adequate protection against freezing. Refer to Operation and Maintenance Manual, “Fluid Recommendations” for more information.

If an engine is out of operation and if use of the engine is not planned, special precautions should be made. If the engine will be stored for more than 1 month, a complete protection procedure is recommended.

Ensure that the outlets on the Clean Emission Module (CEM) are capped. Ensure that the Diesel Exhaust Fluid (DEF) connections are capped.

To prevent damage to the exhaust outlet connection during storage, the weight of the CEM must not act on the exhaust outlet.

For more detailed information on engine storage, refer to Special Instruction, SEHS9031, “Storage Procedure For Caterpillar Products”.

Your Caterpillar dealer can help in preparing the engine for extended storage periods.

Aftertreatment

The engine must be allowed to perform a DEF purge before the battery disconnect switch is turned off. Allow 2 minutes after the engine has stopped before disconnecting the battery disconnect switch.

The exhaust outlet of the aftertreatment must be capped. To prevent damage to the exhaust outlet connection during storage, the weight of the CEM must not act on the exhaust outlet.

DEF Tank Storage

1. Ensure normal engine shutdown, allow the DEF to be purged. Do not disconnect the battery disconnect switch, allow 2 minutes after key off before disconnection.

2. Fill the tank with DEF that meets all the requirement defined in ISO 22241-1.

3. Ensure that all DEF lines and electrical connection are connected prior to prevent crystal from forming.

4. Ensure that the DEF filler cap is correctly installed.

Removal from Storage

DEF has a limited life, refer to table 3 for the time and temperature range. DEF that is outside this range MUST be replaced.

On removal from storage the DEF quality in the tank must be tested with a refractometer. The DEF in the tank must meet the requirements defined in ISO 22241-1 and comply with table 3 .

1. If necessary, drain the tank and fill with DEF that meets ISO 22241-1.

2. Replace the DEF filter. Refer to Operation and Maintenance Manual, “Diesel Exhaust Fluid Filter (Emission Related Component) - Replace” for the correct procedure.

3. Ensure that the drive belt is correctly installed. Ensure that all engine coolant and engine oil has the correct specification and grade. Ensure that the coolant and the engine oil are at the correct level. Start the engine. If a fault becomes active turn off the engine, allow 2 minutes for the DEF system to purge, then restart the engine.

4. If the fault continues to stay active, refer to Troubleshooting for more information.

Table 3

Temperature	Duration
10° C (50° F)	36 months
25° C (77° F)	18 months
30° C (86° F)	12 months
35° C (95° F)(1)	6 months

(1) At 35° C, significant degradation can occur. Check every batch before use.

Features and Controls

i07925952

Alarms and Shutoffs

SMCS Code: 7400

Shutoffs

The shutoffs are electrically operated or mechanically operated. The electrically operated shutoffs are controlled by the ECM.

Shutoffs are set at critical levels for the following items:

- Operating temperature
- Operating pressure
- Operating level
- Operating rpm

The particular shutoff may need to be reset before the engine will start.

NOTICE
Always determine the cause of the engine shutdown. Make necessary repairs before attempting to restart the engine.

Be familiar with the following items:

- Types and locations of shutoff
- Conditions which cause each shutoff to function
- The resetting procedure that is required to restart the engine

Alarms

The alarms are electrically operated. The operations of the alarms are controlled by the ECM.

The alarm is operated by a sensor or by a switch. When the sensor or the switch is activated, a signal is sent to the ECM. An event code is created by the ECM. The ECM will send a signal to illuminate the lamp.

Your engine may be equipped with the following sensors or switches:

Coolant temperature – The coolant temperature sensor indicates high jacket water coolant temperature.

Intake manifold air temperature – The intake manifold air temperature sensor indicates high intake air temperature.

Intake manifold pressure – The intake manifold pressure sensor checks the rated pressure in the engine manifold.

Fuel rail pressure – The fuel rail pressure sensor measures the high pressure or low pressure in the fuel rail. The ECM will Check the pressure.

Engine oil pressure – The engine oil pressure switch indicates when oil pressure drops below rated system pressure, at a set engine speed.

Engine overspeed – If the engine rpm exceeds the overspeed setting, the alarm will be activated.

Air filter restriction – The switch checks the air filter when the engine is operating.

User-defined switch – This switch can shut down the engine remotely.

Water in fuel switch – This switch checks for water in the primary fuel filter when the engine is operating.

Fuel temperature – The fuel temperature sensor monitors the pressurized fuel in the high-pressure fuel pump.

NRS absolute pressure and NRS delta pressure – A combined pressure sensor across the NRS valve.

Note: The sensing element of the coolant temperature sensor must be submerged in coolant to operate.

Engines may be equipped with alarms to alert the operator when undesirable operating conditions occur.

NOTICE
When an alarm is activated, corrective measures must be taken before the situation becomes an emergency in order to avoid possible engine damage.

If corrective measures are not taken within a reasonable time, engine damage could result. The alarm will continue until the condition is corrected. The alarm may need to be reset.

Note: If installed, the coolant level switch and the oil level switch are indicators. Both switches operate when the application is on level ground and the engine RPM at zero.

Aftertreatment System

- Diesel Oxidation Catalyst (DOC)
- Selective Catalyst Reduction (SCR)
- Diesel Particulate Filter (DPF)

Soot Sensors – The soot sensor monitor the soot level within the DPF.

NOx Sensors – Two NOx sensors monitor the NOx concentration within the exhaust gas before and after the selective catalyst reduction module.

Temperature Sensor – A temperature sensor after the engine exhaust gas exit, after the DOC and before the SCR module monitor the temperatures within the system.

DEF Tank Header Unit – The DEF level sensor monitors the volume of fluid in the tank and signals the ECM if the level drops below a given point.

Testing

Turning the keyswitch to the ON position will check the indicator lights on the control panel. All the indicator lights will be illuminated for 2 seconds after the keyswitch is operated. Replace suspect bulbs immediately.

Refer to Troubleshooting for more information.

i05408886

Battery Disconnect Switch (If Equipped)

SMCS Code: 1411

Allow at least 2 minutes after the engine has stopped before you turn the battery disconnect switch to OFF. Disconnecting the battery power too soon will prevent purging of the Diesel Exhaust Fluid (DEF) lines after the engine is shut down. Also, during the 2 minutes the engine Electronic Control Module (ECM) is active storing information from the engine and aftertreatment sensors.

Not allowing the DEF purge to be performed can damage the DEF system. Not allowing the engine ECM time to store the information from the sensors can damage to emission control system.

Some applications, the engine can be equipped with a wait to disconnect lamp. The wait to disconnect lamp will be illuminated during engine operation and will be extinguished approximately 2 minutes after the engine has stopped.

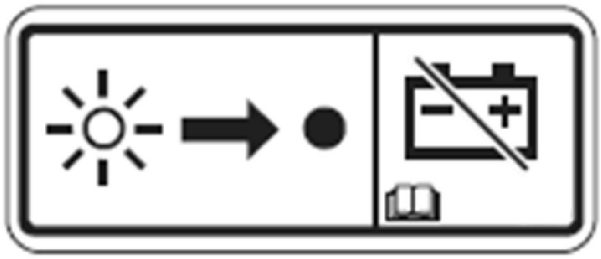


Illustration 50 g03265058

Battery disconnect switch label

NOTICE
Do not turn off the battery disconnect switch until the indicator lamp has turned off. If the switch is turned off when the indicator lamp is illuminated the Diesel Exhaust Fluid (DEF) system will not purge the DEF. If the DEF does not purge, DEF could freeze and damage the pump and lines.

NOTICE
Never move the battery disconnect switch to the OFF position while the engine is operating. Serious damage to the electrical system could result.

i08773514

Gauges and Indicators

SMCS Code: 7450

Your engine may not have the same gauges or all the gauges that are described. For more information about the gauge package, see the OEM information.

Gauges provide indications of engine performance. Ensure that the gauges are in good working order. Determine the normal operating range by observing the gauges over a period.

Noticeable changes in gauge readings indicate potential gauge or engine problems. Problems may also be indicated by gauge readings that change even if the readings are within specifications. Determine and correct the cause of any significant change in the readings. Consult your Caterpillar dealer for assistance.

Some engine applications are equipped with Indicator Lamps. Indicator lamps can be used as a diagnostic aid. There are two lamps. One lamp has an orange lens and the other lamp has a red lens.

These indicator lamps can be used in two ways:

- The indicator lamps can be used to identify the current operational status of the engine. The indicator lamps can also indicate that the engine has a fault. This system is automatically operated via the ignition switch.
- The indicator lamps can be used to identify active diagnostic codes. This system is activated by pressing the Flash Code button.

Refer to the Troubleshooting Guide, “Indicator Lamps” for further information.

NOTICE
If no oil pressure is indicated, STOP the engine. If maximum coolant temperature is exceeded, STOP the engine. Engine damage can result.



Engine Oil Pressure – The oil pressure should be greatest after a cold engine is started. The typical engine oil pressure with SAE10W40 is 350 to 450 kPa (50 to 65 psi) at rated rpm.

A lower oil pressure is normal at low idle. If the engine speed and load are stable and the gauge reading changes, perform the following procedure:

1. Remove the load.

2. Stop the engine.

3. Check and maintain the oil level.



Jacket Water Coolant Temperature – Typical temperature range is 82° to 94°C (179.6° to 169.2°F). This temperature range will vary according to engine load and the ambient temperature.

A 100 kPa (14.5 psi) radiator cap must be installed on the cooling system. The maximum temperature for the cooling system is 108° C (226.4° F). This temperature is measured at the outlet for the water temperature regulator. The engine coolant temperature is monitored by the engine sensors and the engine ECM. This programming cannot be altered. Derates can occur if the maximum engine coolant temperature is exceeded.

If the engine is operating above the normal range, reduce the engine load. If high coolant temperatures are a frequent event, perform the following procedures:

1. Reduce the load on the engine.

2. Determine if the engine must be shut down immediately or if the engine can be cooled by reducing the load.

3. Inspect the cooling system for leaks. If necessary, consult your Caterpillar dealer for assistance.



Tachometer – This gauge indicates engine speed (rpm). When the throttle control lever is moved to the full throttle position without load, the engine is running at high idle. The engine is running at the full load rpm when the throttle control lever is at the full throttle position with maximum rated load.

NOTICE
To help prevent engine damage, never exceed the high idle rpm. Overspeeding can result in serious damage to the engine. Operation at speeds exceeding high idle rpm should be kept to a minimum.



Ammeter – This gauge indicates the amount of charge or discharge in the battery charging circuit. Operation of the indicator should be to the “+” side of “0” (zero).



Engine - Electrical Preheat – This indicator comes “ON” to show that the glow plug preheat is active. The engine should be set to “RUN” only after indicator lamp on the display turns “OFF” .



Fuel Level – This gauge indicates the fuel level in the fuel tank. The fuel level gauge operates when the “START/STOP” switch is in the “on” position.



Service Hour Meter – The gauge indicates total operating hours of the engine.

Indicator Lamps

- Shutdown lamp
- Warning lamp
- Low oil pressure lamp
- Wait to start lamp (Glow plug warning lamp)

For information, refer to this manual, “Monitoring System (Table for the Indicator Lamps)” for the sequence of operation of the shutdown lamp and the warning lamp.

The function of the wait to start lamp is automatically controlled at engine start-up.

The function of the low oil pressure lamp is controlled by the engine ECM. If low oil pressure is detected, the lamp will be illuminated. The reason for the illumination of the low-pressure lamp should be investigated immediately.

The glow plug warning lamp will flash to show that the engine is being held at low speed. This function will be performed at engine starting and the duration will depend on ambient temperature and engine temperature.

The lamps will illuminate for 2 seconds to check that the lamps are functioning when the keyswitch is turned to the ON position. If any of the lamps stay illuminated, or a lamp fails to be illuminated the reason should be investigated immediately.

Aftertreatment Lamps and gauges

All applications will require the following lamps and gauges:

- Emission malfunction lamp
- Action indicator lamp
- Gauge for Diesel Exhaust Fluid (DEF)
- Low warning lamp for DEF
- Wait to disconnect lamp (optional)

The wait to disconnect lamp will be illuminated during engine operation and will be extinguished approximately 2 minutes after the engine has stopped. Do not disconnect the battery disconnect switch during the period the lamp is illuminated. The diesel exhaust fluid system will be purged during this time. Also, during the 2 minutes the engine electronic control module is active storing information from the engine and aftertreatment sensors.

Note: The wait to disconnect lamp will not be checked at key on.

i07682718

Monitoring System (Engine Indicators and Aftertreatment Indicators)

SMCS Code: 1900; 7400; 7450; 7451

Engine Indicator Lamps

Note: When in operation the amber warning lamp has three states, solid, flashing, and fast flashing. The sequence is to give a visual indication of the importance of the warning. Some application can have an audible warning installed.

Ensure that the engine maintenance is carried out at the correct intervals. A lack of maintenance can result in illumination of the warning lamp. For the correct intervals of maintenance, refer to the Operation and Maintenance Manual, “Maintenance Interval Schedule”.

Table 4

Indicator Lamp Table				
Warning Lamp	**Shutdown Action Indicator Lamp**	**Lamp State**	**Description of the Indication**	**Engine Status**
On	On	Lamp Check	When the keyswitch is moved to the ON position, the lamps come on for 2 seconds and the lamps will then go off.	The keyswitch is in the ON position but the engine has not yet been cranked.
If any of the indicators will not illuminate during indicator check, the fault must be investigated immediately If any Indicators stay illuminated or flash, the fault must be investigated immediately.				
Off	Off	No Faults	With the engine in operation, there are no active warnings, diagnostic codes, or event codes.	The engine is operating with no detected faults.
On Solid	Off	Warning	Level 1 warning	The engine is operating normally but there is one or more faults with the electronic management system for the engine.
As soon as possible the fault should be investigated.				
Flashing	Off	Warning	Level 2 warning	The engine continues to be operated, but the level of importance of the warning has increased. Depending on the particular fault and the severity the engine may be de-rated. The engine could be damaged if continued to be operated.
Stop the engine. Investigate the code.				
Flashing	On	Engine Shutdown	Level 3 warning If both the warning lamp and the shutdown lamp are in operation, this issue indicates one of the following conditions. 1. One or more of the shutdown values for the engine protection strategy has been exceeded. 2. A serious active diagnostic code has been detected. If installed, the audible warning will sound. After a short time period, the engine may shut down.	The engine is either shutdown or an engine shutdown is imminent. One or more monitored engine parameters have exceeded the limit for an engine shutdown. This pattern of lamps can be caused by the detection of a serious active diagnostic code. Contact your Caterpillar dealer.

Flash Codes

Some applications may support flash codes. A flash code can be viewed by an indicator lamp that when asked will flash in a particular sequence. The indicator lamp used to view the codes is the warning lamp, the lamp can then be referred to as a diagnostic lamp. For more information refer to this Operation and Maintenance Manual, “Diagnostic Lamp”.

Aftertreatment System

The aftertreatment indicators should illuminate at key on for 2 seconds to test the system. If any of the indicators do not illuminate, the fault must be investigated immediately.

For more information on the illumination of the aftertreatment indicators refer to this Operation and Maintenance Manual, “Selective Catalytic Reduction Warning System”.

i08068136

Sensors and Electrical Components

SMCS Code: 1900; 7400

The illustrations within the following sections are typical location of the sensors or electrical components for an industrial engine. Specific engines may appear different due to differences in applications.

NRS – NOx Reduction System

DEF – Diesel Exhaust Fluid

DOC – Diesel Oxidation Catalyst

DPF – Diesel Particulate Filter

SCR – Selective Catalytic Reduction

ECM – Electronic Control Module

NOx – Nitrogen Oxides

C3.6 Industrial Engines

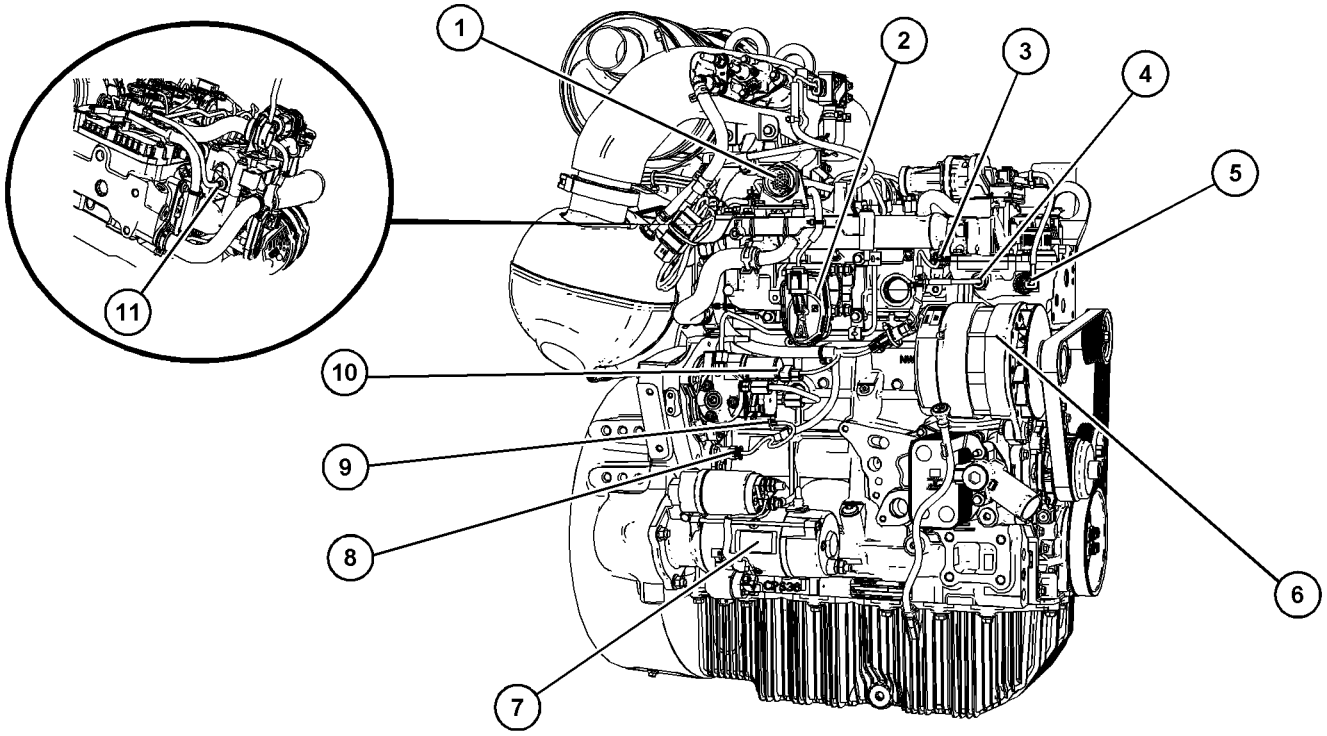


Illustration 51 g06299856

Typical example

(1) Engine 47-pin interface connector
(2) Inlet throttle valve
(3) Inlet manifold combined pressure and temperature sensor
(4) Temperature sensor pre-NRS valve
(5) Coolant temperature sensor
(6) Alternator
(7) Starting motor
(8) Camshaft speed/timing sensor
(9) Fuel temperature sensor
(10) Control valve for high-pressure fuel pump
(11) Temperature sensor post NRS valve

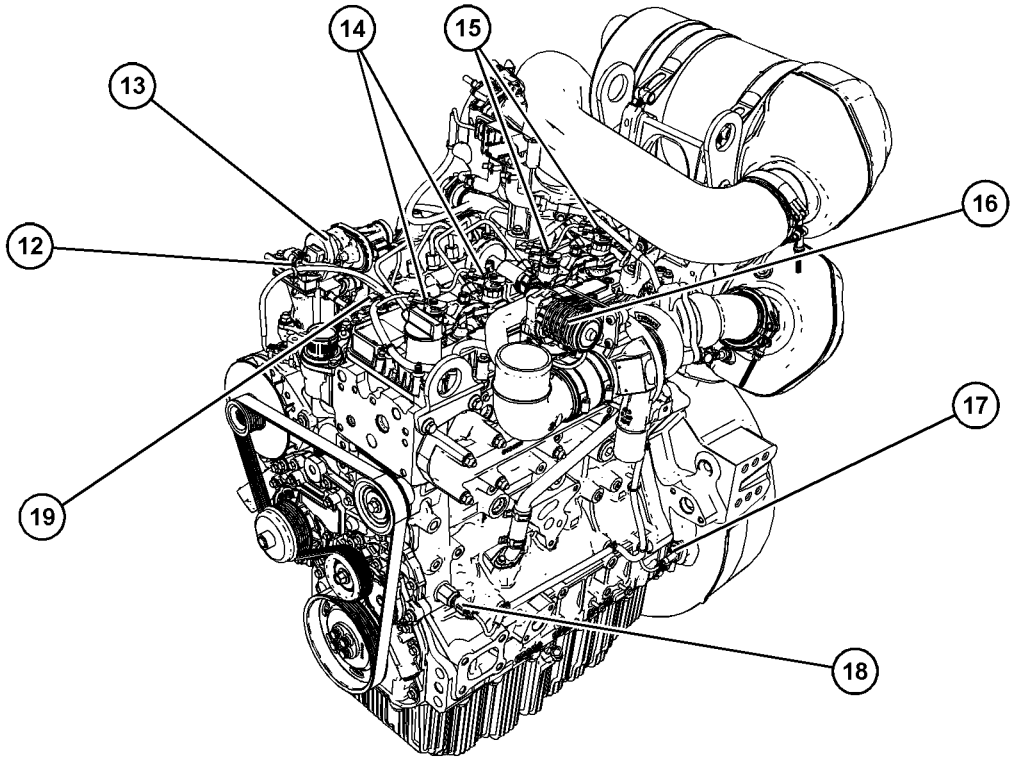


Illustration 52 g06299988

Typical example

(12) NRS Differential pressure sensor
(13) NRS valve
(14) Injector 1 and injector 2
(15) Injector 3 and injector 4
(16) Actuator for turbocharger
(17) Crankshaft speed/timing sensor
(18) Engine oil pressure switch
(19) Fuel rail pressure sensor

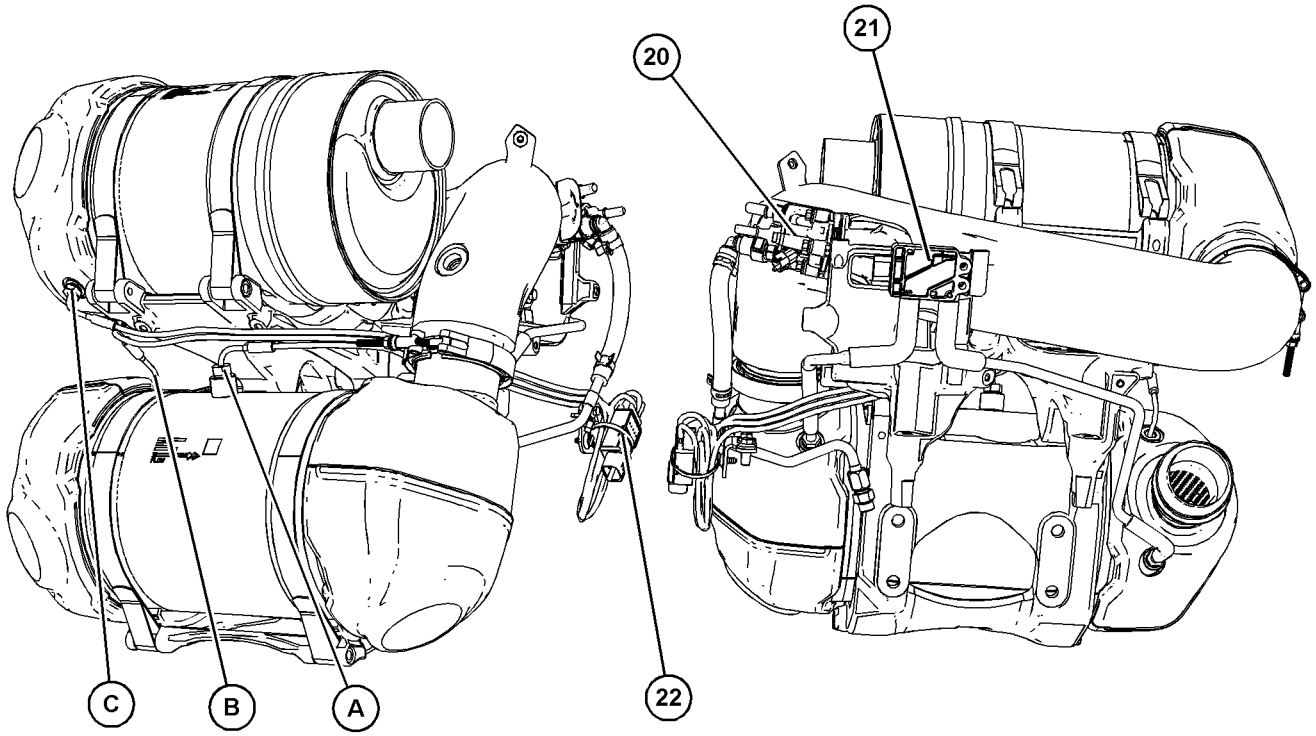


Illustration 53 g06300327

Typical example

(20) DEF Injector
(21) DPF Differential pressure sensor
(22) Temperature sensor controller with 3 probes across aftertreatment (in transport position)
(A) DPF temperature probe
(B) DOC temperature probe
(C) SCR temperature probe

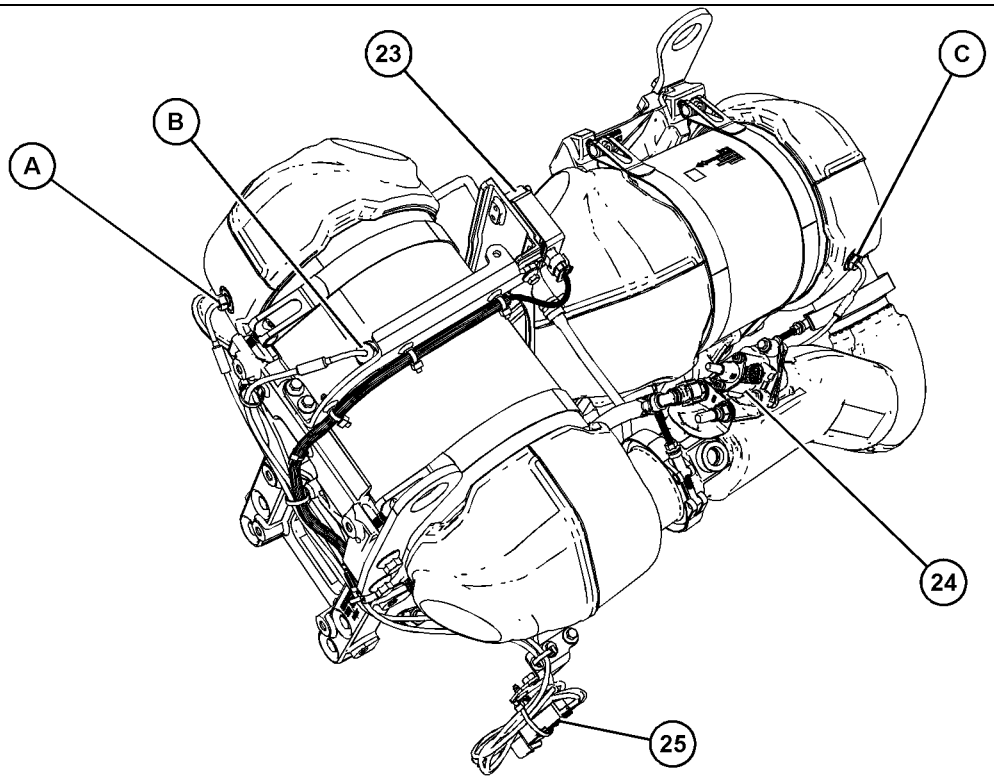


Illustration 54 g06511813

Industrial engine equipped with a balancer

(23) DPF Differential pressure sensor
(24) DEF Injector
(25) Temperature sensor controller connection with 3 probes across aftertreatment
(A) DOC temperature probe
(B) DPF temperature probe
(C) SCR temperature probe

Off Engine Sensors and Electrical Components

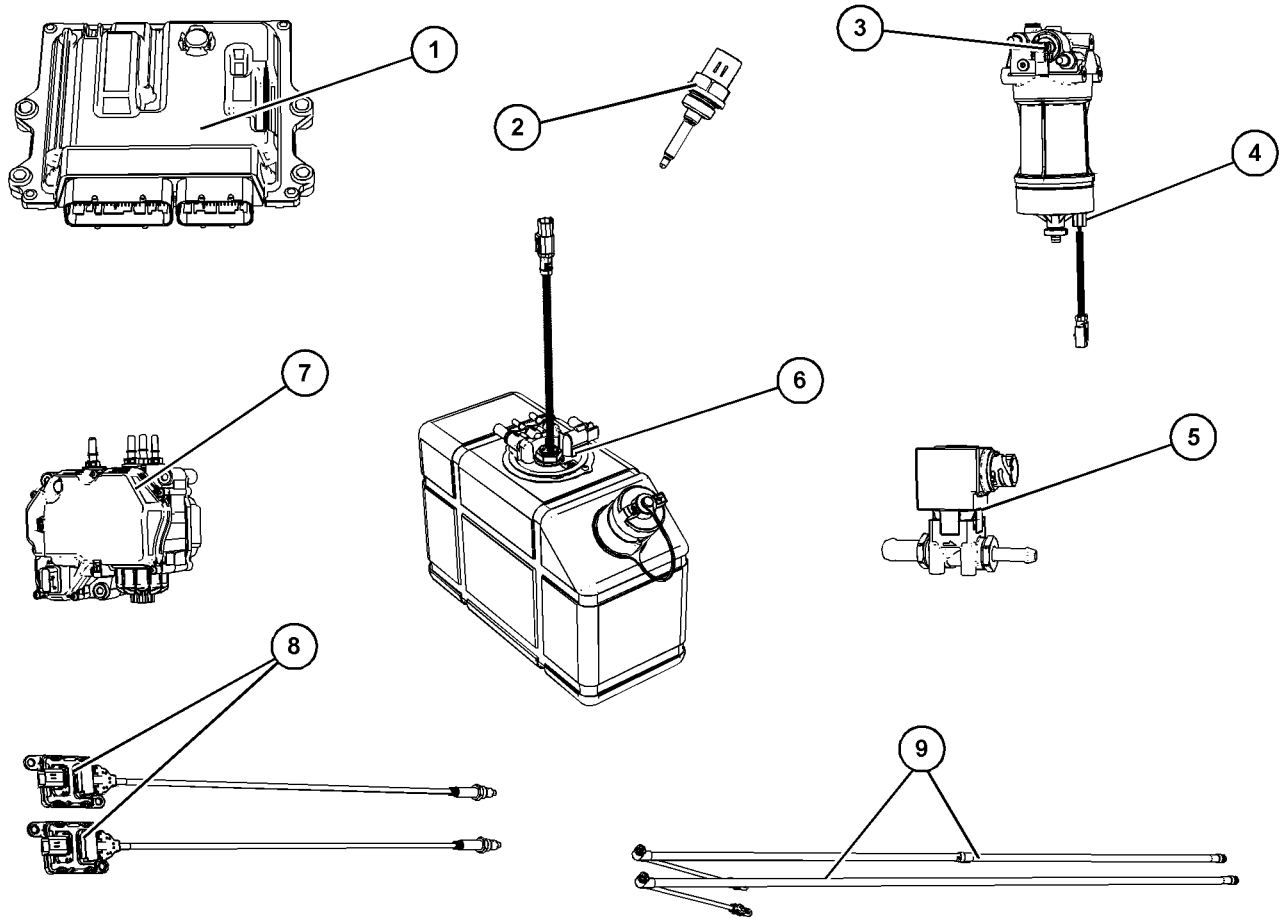


Illustration 55 g06300407

Typical example

(1) ECM
(2) If equipped, Air intake temperature sensor (Air Cleaner)
(3) Priming pump
(4) Water in fuel switch
(5) Coolant diverter valve
(6) DEF tank level, temperature sensor, and DEF quality sensor
(7) DEF pump
(8) NOx sensors
(9) DEF heated lines

C2.8 Industrial Engine

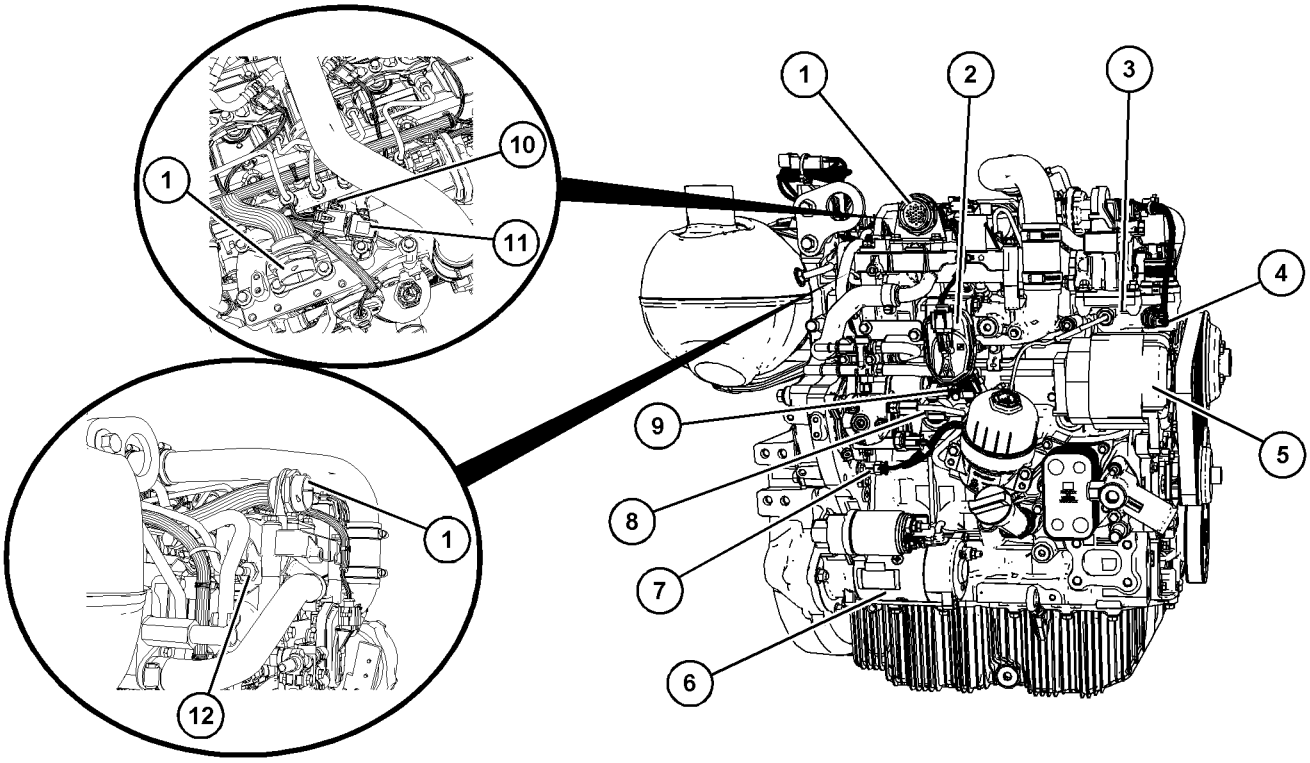


Illustration 56 g06481405

Typical example

(1) Engine 47-pin interface connector
(2) Inlet throttle valve
(3) Temperature sensor pre-NRS valve
(4) Coolant temperature sensor
(5) Alternator
(6) Starting motor
(7) Camshaft speed/timing sensor
(8) Fuel temperature sensor
(9) Control valve for high-pressure fuel pump
(10) Inlet Manifold Temperature Sensor
(11) Inlet Manifold Pressure Sensor
(12) Temperature sensor post NRS valve

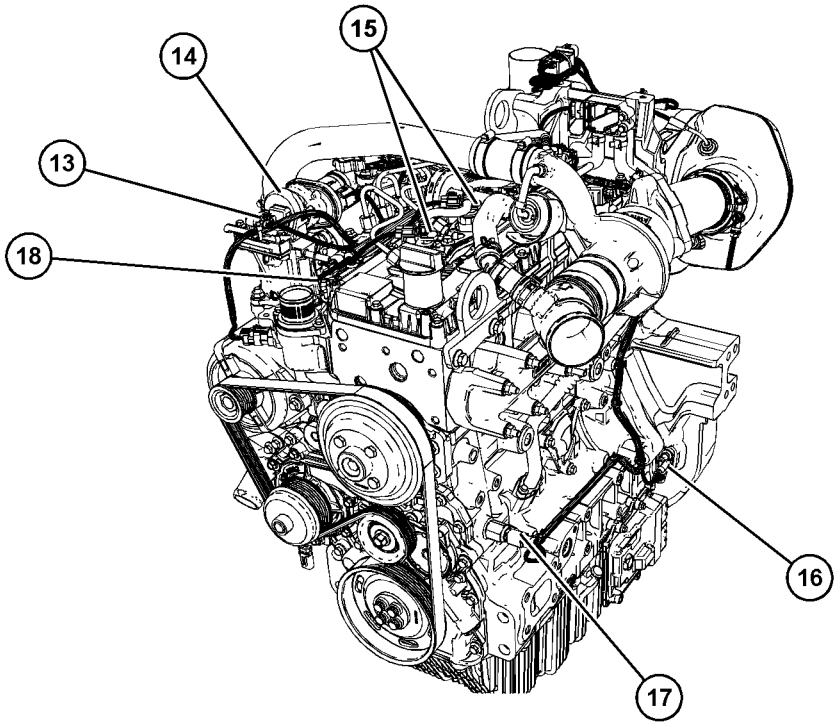


Illustration 57 g06481410

Typical example

(13) NRS Differential pressure sensor
(14) NRS valve
(15) Injector 1 and injector 2 (Injector 3 and injector 4 not shown)
(16) Crankshaft speed/timing sensor
(17) Engine oil pressure switch
(18) Fuel rail pressure sensor

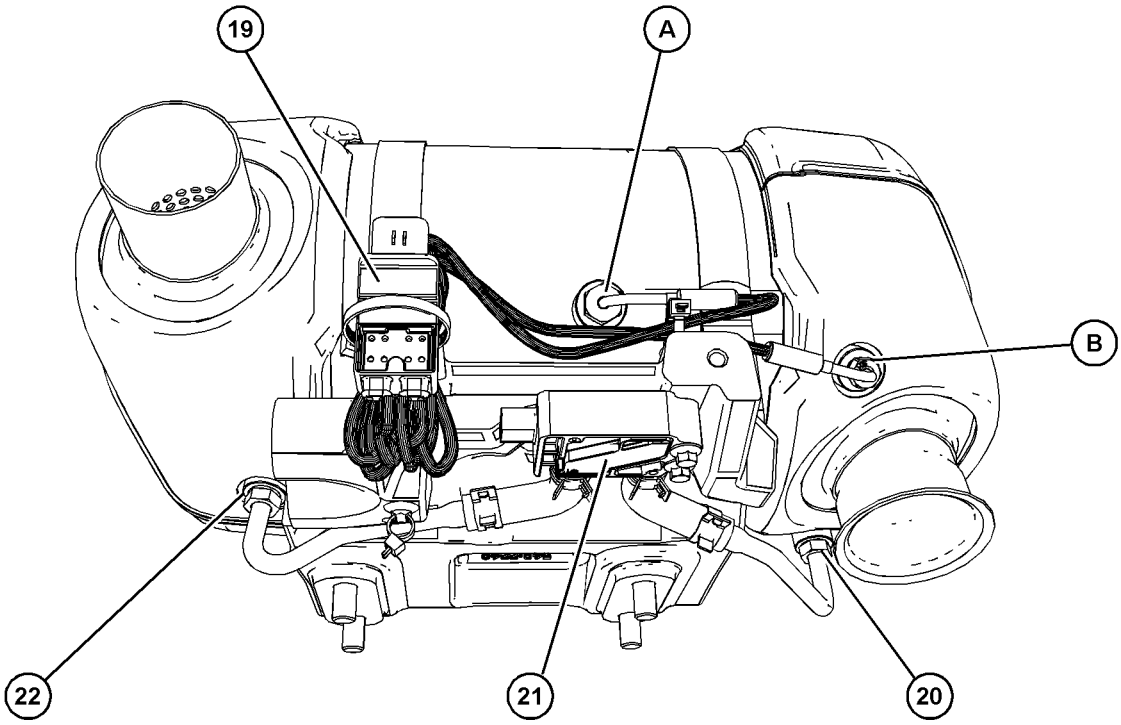


Illustration 58 g06481414

Typical example

(19) Temperature sensor controller with 2 probes across aftertreatment (in transport position)
(20) Tube assembly for Differential pressure
(21) DPF Differential pressure sensor
(22) Tube assembly for Differential pressure
(A) DPF temperature probe
(B) DOC temperature probe

C2.8 Off Engine Sensors and Electrical Components

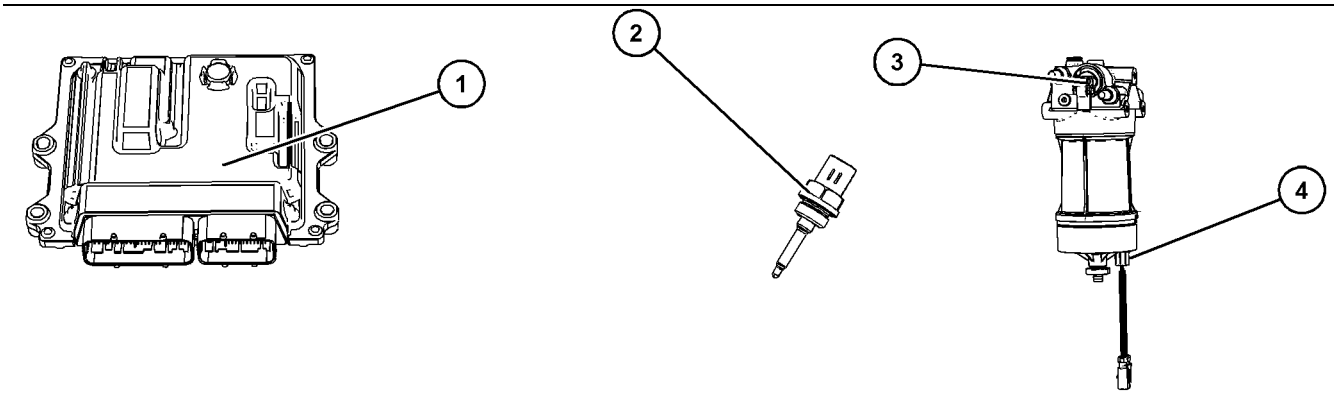


Illustration 59 g06481423

Typical example

(1) ECM
(2) If equipped, Air intake temperature sensor (Air Cleaner)
(3) Priming pump
(4) Water in fuel switch

Engine Diagnostics

i01796959

Self-Diagnostics

SMCS Code: 1000; 1900; 1901; 1902

Caterpillar Electronic Engines have the capability to perform a self-diagnostics test. When the system detects an active problem, a diagnostic lamp is activated. Diagnostic codes will be stored in permanent memory in the Electronic Control Module (ECM). The diagnostic codes can be retrieved by using Caterpillar electronic service tools.

Some installations have electronic displays that provide direct readouts of the engine diagnostic codes. Refer to the manual that is provided by the OEM for more information on retrieving engine diagnostic codes.

Active codes represent problems that currently exist. These problems should be investigated first.

Logged codes represent the following items:

- Intermittent problems
- Recorded events
- Performance history

The problems may have been repaired since the logging of the code. These codes do not indicate that a repair is needed. The codes are guides or signals when a situation exists. Codes may be helpful to troubleshoot problems.

When the problems have been corrected, the corresponding logged fault codes should be cleared.

i03554520

Diagnostic Lamp

SMCS Code: 1000; 1900; 1901; 1902; 7451

A diagnostic lamp is used to indicate the existence of an active fault. A fault diagnostic code will remain active until the problem is repaired. The diagnostic code may be retrieved by using the electronic service tool.

i04907245

Diagnostic Flash Code Retrieval

SMCS Code: 1000; 1900; 1901; 1902

Use the following procedure to retrieve the flash codes if the engine is equipped with a “DIAGNOSTIC” lamp:

1. Move the keyswitch from the on/off two times within 3 seconds.
2. The shutdown warning lamp will flash once.
3. A flashing YELLOW lamp indicates a 3-digit diagnostic code for the engine. The sequence of flashes represents the system diagnostic message. Count the first sequence of flashes in order to determine the first digit of the flash code. After a two second pause, the second sequence of flashes will identify the second digit of the flash code. After the second pause, the third sequence of flashes will identify the flash code.
4. After the diagnostic codes have been displayed, the shutdown lamp will flash twice and the indicator lamp will start to flash the logged diagnostic codes.
5. After the logged diagnostic codes have been displayed, the shutdown lamp will flash three times in order to indication that the code sequences have finished.

Note: If there are no diagnostic codes or logged diagnostic codes, the system will flash the code 551.

i05406659

Fault Logging

SMCS Code: 1000; 1900; 1901; 1902

The system provides the capability of Fault Logging. When the Electronic Control Module (ECM) generates an active diagnostic code, the code will be logged in the memory of the ECM. The codes that have been logged by the ECM can be identified by the electronic service tool. The active codes that have been logged will be cleared when the fault has been rectified or the fault is no longer active. The following logged faults cannot be cleared from the memory of the ECM without using a factory password: Overspeed, low engine oil pressure, high engine coolant temperature, and aftertreatment codes.

i03554534

Engine Operation with Active Diagnostic Codes

SMCS Code: 1000; 1900; 1901; 1902

If a diagnostic lamp illuminates during normal engine operation, the system has identified a situation that is not within the specification. Use electronic service tools to check the active diagnostic codes.

Note: If the customer has selected “DERATE” and if there is a low oil pressure condition, the Electronic Control Module (ECM) will limit the engine power until the problem is corrected. If the oil pressure is within the normal range, the engine may be operated at the rated speed and load. However, maintenance should be performed as soon as possible.

The active diagnostic code should be investigated. The cause of the problem should be corrected as soon as possible. If the cause of the active diagnostic code is repaired and there is only one active diagnostic code, the diagnostic lamp will turn off.

Operation of the engine and performance of the engine can be limited as a result of the active diagnostic code that is generated. Acceleration rates may be significantly slower. Refer to the Troubleshooting Guide for more information on the relationship between these active diagnostic codes and engine performance.

i01797063

Engine Operation with Intermittent Diagnostic Codes

SMCS Code: 1000; 1900; 1901; 1902

If a diagnostic lamp illuminates during normal engine operation and the diagnostic lamp shuts off, an intermittent fault may have occurred. If a fault has occurred, the fault will be logged into the memory of the Electronic Control Module (ECM).

In most cases, it is not necessary to stop the engine because of an intermittent code. However, the operator should retrieve the logged fault codes and the operator should reference the appropriate information in order to identify the nature of the event. The operator should log any observation that could have caused the lamp to light.

- Low power
- Limits of the engine speed
- Excessive smoke, etc

This information can be useful to help troubleshoot the situation. The information can also be used for future reference. For more information on diagnostic codes, refer to the Troubleshooting Guide for this engine.

Engine Starting

i08171598

Before Starting Engine

SMCS Code: 1000; 1400; 1450

Perform the required daily maintenance and other periodic maintenance before the engine is started. Inspect the engine compartment. This inspection can help prevent major repairs later. Refer to the Operation and Maintenance Manual, “Maintenance Interval Schedule” for more information.

- Ensure that the engine has an adequate fuel supply.
- Ensure that the engine has an adequate diesel exhaust fluid supply
- Open the fuel supply valve (if equipped).
- Drain the water separators.

NOTICE
All valves in the fuel return line must be open and fuel supply lines must be open. Damage to the fuel system can occur if fuel lines are closed with the engine in operation.

If the engine has not been started for several weeks, fuel may have drained from the fuel system. Air may have entered the filter housing. Also, when fuel filters have been changed, some air pockets will be trapped in the engine. In these instances, prime the fuel system. Refer to the Operation and Maintenance Manual, “Fuel System - Prime” for more information on priming the fuel system. Also, check that the fuel specification is correct and that the fuel condition is correct. Refer to the Operation and Maintenance Manual, “Fuel Recommendations”.

⚠ WARNING

Engine exhaust contains products of combustion which may be harmful to your health. Always start and operate the engine in a well ventilated area and, if in an enclosed area, vent the exhaust to the outside.

- Do not start the engine or move any of the controls if there is a “DO NOT OPERATE” warning tag or similar warning tag attached to the start switch or to the controls.
- Reset all the shutoffs or alarm components.
- Ensure that any driven equipment has been disengaged. Minimize electrical loads or remove any electrical loads.

i08165229

Cold Weather Starting

SMCS Code: 1000; 1250; 1450; 1453; 1456; 1900

⚠ WARNING

Do not use aerosol types of starting aids such as ether. Such use could result in an explosion and personal injury.

The ability to start the engine will be improved at temperatures below −18 °C (0 °F) from the use of a jacket water heater or extra battery capacity.

The following items provide a means of minimizing starting problems and fuel problems in cold weather: Engine oil pan heaters, jacket water heaters, fuel heaters, and fuel line insulation.

Use the procedure that follows for cold weather starting.

Note: Do not adjust the engine speed control during start-up. The electronic control module (ECM) will control the engine speed during start-up.

1. Disengage any driven equipment.

Note: During key ON, the indicator lamps will be illuminated for 2 seconds to check the lamp operation. If any of the indicator lamps do not illuminate check the bulb. If any indicator lamps stay illuminated or flash, refer to Troubleshooting, “Indicator Lamp Circuit - Test”.

2. Turn the keyswitch to the RUN position. Leave the keyswitch in the RUN position until the warning light for the glow plugs is extinguished.

3. When the warning light for the glow plugs is extinguished, turn the keyswitch to the START position to engage the electric starting motor and crank the engine.

Note: The operating period of the warning light for the glow plugs will change due to the ambient air temperature.

NOTICE
Do not engage the starting motor when flywheel is turning. Do not start the engine under load.

If the engine fails to start within 30 seconds, release the starter switch or button and wait two minutes to allow the starting motor to cool before attempting to start the engine again.

4. Allow the keyswitch to return to the RUN position after the engine starts.

5. Repeat step 2 through step 4 if the engine fails to start.

Note: After starting, the engine may be held at low speed for a duration between 1 and 60 seconds to allow engine systems to stabilize. The duration will depend on ambient temperature, time since last run and other factors. The glow plug warning lamp will flash to indicate that the engine has been held at low speed.

6. The engine should not be raced to speed up the warm-up process. Allow the engine to idle for 3 to 5 minutes, or allow the engine to idle until the water temperature indicator begins to rise. When idling after the engine has started in cold weather, increase the engine rpm from low idle to 1200 rpm. This operation will warm up the engine more quickly.

7. Operate the engine at low load until all systems reach operating temperature. Check the gauges during the warm-up period.

Following a cold start, white vapor can be seen from the tailpipe. This vapor is normal and caused by condensation leaving the exhaust system after warming up. The white vapor should clear following working of the engine.

i08165241

Starting the Engine

SMCS Code: 1000; 1450

Note: Do not adjust the engine speed control during start-up. The electronic control module (ECM) will control the engine speed during start-up.

Starting the Engine

1. Disengage any equipment that is driven by the engine.

2. Turn the keyswitch to the RUN position. Leave the keyswitch in the RUN position until the warning light for the glow plugs is extinguished.

Note: If the keyswitch has been left in the RUN position for a long period of time without engaging the electric starting motor, turn the keyswitch from the RUN position to the OFF position and then back to the RUN position to reactivate the glow plug preheat stage.

Note: During the key on, the indicator lamps will be illuminated for 2 seconds to check lamp operation. If any of the lamps do not illuminate, check the bulb. If any indicator lamps stay illuminated or flash, refer to Troubleshooting, “Indicator Lamp Circuit - Test”.

3. When the warning light for the glow plugs is extinguished, turn the keyswitch to the START position to engage the electric starting motor and crank the engine.

Note: The operating period of the warning light for the glow plugs will change due to the temperature of the engine.

NOTICE
Do not engage the starting motor when flywheel is turning. Do not start the engine under load.

If the engine fails to start within 30 seconds, release the starter switch or button and wait two minutes to allow the starting motor to cool before attempting to start the engine again.

4. Allow the keyswitch to return to the RUN position after the engine starts.

5. Repeat step 2 through step 4 if the engine fails to start.

6. After starting, the engine may be held at low speed for a duration between 1 and 60 seconds to allow engine systems to stabilize. The duration will depend on ambient temperature, time since last run and other factors. The glow plug warning light will flash to indicate that the engine is being held at low speed in some applications.

i07926219

Starting with Jump Start Cables

(Do Not Use This Procedure in Hazardous Locations that have Explosive Atmospheres)

SMCS Code: 1000; 1401; 1402; 1900

⚠ WARNING

The connection of battery cables to a battery and the disconnection of battery cables from a battery may cause an explosion which may result in injury or death. The connection and the disconnection of other electrical equipment may also cause an explosion which may result in injury or death. The procedures for the connection and the disconnection of battery cables and other electrical equipment should only be performed in a nonexplosive atmosphere.

⚠ WARNING

Improper jump start cable connections can cause an explosion resulting in personal injury.

Prevent sparks near the batteries. Sparks could cause vapors to explode. Do not allow jump start cable ends to contact each other or the engine.

Note: If possible, first diagnose the reason for the starting failure. Refer to Troubleshooting, “Engine Will Not Crank and Engine Cranks But Will Not Start” for further information. Make any necessary repairs. If the engine will not start only due to the condition of the battery, either charge the battery, or start the engine by using another battery with jump-start cables. The condition of the battery can be rechecked after the engine has been switched OFF.

NOTICE
For C3.6 industrial engines, ensure that a 12 VDC or 24 VDC battery source is used to start the engine. For C2.8 industrial engines, use a 12 VDC battery source to start the engine. Never attempt to start an engine from an external power source such as electric welding equipment, which has a voltage that is unsuitable for engine starting and will damage the electrical system.

NOTICE
Using a battery source with the same voltage as the electric starting motor. Use ONLY equal voltage for jump starting. The use of higher voltage will damage the electrical system.

Do not reverse the battery cables. The alternator can be damaged. Attach ground cable last and remove first.

Turn all electrical accessories OFF before attaching the jump start cables.

Ensure that the main power switch is in the OFF position before attaching the jump start cables to the engine being started.

1. Turn the start switch on the stalled engine to the OFF position. Turn off all the engines accessories.
2. Connect one positive end of the jump-start cable to the positive cable terminal of the discharged battery. Connect the other positive end of the jump-start cable to the positive cable terminal of the electrical source.
3. Connect one negative end of the jump-start cable to the negative cable terminal of the electrical source. Connect the other negative end of the jump-start cable to the engine block or to the chassis ground. This procedure helps to prevent potential sparks from igniting the combustible gases that are produced by some batteries.

Note: The engine ECM must be powered before the starting motor is operated or damage can occur.

4. Start the engine in the normal operating procedure. Refer to this Operation and Maintenance Manual, “Starting the Engine”.
5. Immediately after the engine is started, disconnect the jump-start cables in reverse order.

After jump starting, the alternator may not be able to recharge fully batteries that are severely discharged. The batteries must be replaced or charged to the proper voltage with a battery charger after the engine is stopped. Many batteries which are considered unusable are still rechargeable. Refer to Operation and Maintenance Manual, “Battery - Replace” and Testing and Adjusting Manual, “Battery - Test”.

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After Starting Engine

SMCS Code: 1000

After starting, the engine may be held at low speed for a duration between 1 and 60 seconds to allow engine systems to stabilize. Holding the engine speed low is controlled by the Electronic Control Module (ECM). The duration will depend on ambient temperature, time since last run and other factors.

Note: In ambient temperatures from 15° to 20°C (59° to 68°F), the warm-up time is approximately 20 to 30 minutes. In temperatures below 15°C (59°F), additional warm-up time may be required. In temperatures greater than 20° C (68° F), warm-up time may be less.

When the engine idles during warm-up, observe the following conditions:

Do not check the high-pressure fuel lines with the engine or the starting motor in operation. If you inspect the engine in operation, always use the correct inspection procedure to avoid a fluid penetration hazard. Refer to Operation and Maintenance Manual, “General hazard Information”.

- Check for any fluid or air leaks at idle rpm and at one-half full rpm (no load on the engine), before operating the engine under load. This check may not be possible in some applications.

- Allow the engine to idle for 3 to 5 minutes, or allow the engine to idle until the water temperature indicator begins to rise. Check all gauges during the warm-up period.

Constant speed engines should be allowed to operate at low idle for 3 minutes before used at operational speed. If the low idle option is not available, then operate the engine at operational speed with no load for 2 minutes.

Note: Gauge readings should be observed and the data should be recorded frequently while the engine is operating. Comparing the data over time will help to determine normal readings for each gauge. Comparing data over time will also help detect abnormal operating developments. Significant changes in the readings should be investigated.

Engine Operation

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Engine Operation

SMCS Code: 1000

Correct operation and maintenance are key factors in obtaining the maximum life and economy of the engine. If the directions in the Operation and Maintenance Manual are followed, costs can be minimized and engine service life can be maximized.

The time that is needed for the engine to reach normal operating temperature can be less than the time taken for a walk-around inspection of the engine.

The engine can be operated at the rated rpm after the engine is started and after the engine reaches operating temperature. The engine will reach normal operating temperature sooner during a low engine speed (rpm) and during a low-power demand. This procedure is more effective than idling the engine at no load. The engine should reach operating temperature in a few minutes.

Avoid excess idling. Excessive idling causes carbon buildup, engine slobber, and soot loading of the Diesel Particulate Filter (DPF). These issues are harmful to the engine.

Gauge readings should be observed and the data should be recorded frequently while the engine is operating. Comparing the data over time will help to determine normal readings for each gauge. Comparing data over time will also help detect abnormal operating developments. Significant changes in the readings should be investigated.

Engine Operation and the Aftertreatment System

The exhaust gases and hydrocarbon particles from the engine first pass through Diesel Oxidation Catalyst (DOC). Some of the gasses and matter are oxidized as the gasses pass through the DOC. The gasses then pass through the Diesel Particulate Filter (DPF). The DPF collects the soot and any ash that is produced by the combustion in the engine. During regeneration, the soot is converted into a gas and the ash remains in the DPF. The gasses finally pass through the Selective Catalytic Reduction (SCR). Before the gasses pass through the SCR, Diesel Exhaust Fluid (DEF) is injected into the gas stream.

The DEF is stored in the DEF tank and is pumped to the DEF injector. The DEF injector is controlled by the ECM. The mixtures of DEF and the exhaust gas pass through the SCR reducing the NOx in the exhaust emissions.

The DPF may require the exhaust gas temperature to rise to remove the soot. If necessary, the throttle valve is operated to help in rising the exhaust temperature.

The engine software will control the amount of DEF that will be required to keep the exhaust emission compliant.

This design of DPF will not require a service maintenance interval. The DPF can be expected to function properly for the useful life of the engine (emissions durability period), as defined by regulation, subject to prescribed maintenance requirements being followed.

A fault code will be active for any DPF system-related issue. Follow the troubleshooting guide to rectify the issue.

If the DPF loses function, or is tampered with in any way, the check engine lamp, and an amber action (if equipped) will illuminate. A fault code will also annunciate. The lamps and fault code will remain active until the problem is rectified.

NOTICE
The engine and emissions control system shall be operated, used, and maintained in accordance with the instructions provided. Failure to follow the instructions could result in emissions performance that does not meet the requirements applicable to the category of the engine. No deliberate tampering with, or misuse of the engine emissions control system should take place. Prompt action is critical to rectify any incorrect operation, use, or maintenance of the emissions control system.

Carbon Dioxide (CO_2) Emissions Statement

Emissions regulations require that the value of the CO_2 emissions be reported to the end user.

C3.6 Industrial Engine

For engines with a power rating of 56 kW (75.1 hp) and below, 903.75 g/kWh was determined to be the CO_2 value during the EU type approval process. For engines with a power rating of 70 kW (93.9 hp) and above, 733.73 g/kWh was determined to be the CO_2 value during the EU type approval process. This value was recorded in EU type approval certificate. This CO_2 measurement results from testing over a fixed test cycle, under laboratory conditions, with a parent engine representative of the engine family. This value shall not imply or express any guarantee of the performance of a particular engine.

C2.8 Industrial Engine

856.35 g/kWh was determined to be the CO_2 value during the EU type approval process. This value was recorded in EU type approval certificate. This CO_2 measurement results from testing over a fixed test cycle, under laboratory conditions, with a parent engine representative of the engine family. This value shall not imply or express any guarantee of the performance of a particular engine.

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Engaging the Driven Equipment

SMCS Code: 1000

1. Operate the engine at one-half of the rated rpm, when possible.
2. Engage the driven equipment without a load on the equipment, when possible.

   Interrupted starts put excessive stress on the drive train. Interrupted starts also waste fuel. To get the driven equipment in motion, engage the clutch smoothly with no load on the equipment. This method should produce a start that is smooth and easy. The engine rpm should not increase and the clutch should not slip.
3. Ensure that the ranges of the gauges are normal when the engine is operating at one-half of the rated rpm. Ensure that all gauges operate properly.
4. Increase the engine rpm to the rated rpm. Always increase the engine rpm to the rated rpm before the load is applied.
5. Apply the load. Begin operating the engine at low load. Check the gauges and equipment for proper operation. After normal oil pressure is reached and the temperature gauge begins to move, the engine may be operated at full load. Check the gauges and equipment frequently when the engine is operated under load.

   Extended operation at low idle or at reduced load may cause increased oil consumption and carbon buildup in the cylinders. This carbon buildup results in a loss of power and/or poor performance.

i04018250

Fuel Conservation Practices

SMCS Code: 1000; 1250

The efficiency of the engine can affect the fuel economy. Caterpillar's design and technology in manufacturing provides maximum fuel efficiency in all applications. Follow the recommended procedures in order to attain optimum performance for the life of the engine.

- Avoid spilling fuel.

Fuel expands when the fuel is warmed up. The fuel may overflow from the fuel tank. Inspect fuel lines for leaks. Repair the fuel lines, as needed.

- Be aware of the properties of the different fuels. Use only the recommended fuels. Refer to the Operations and Maintenance Manual, “Fuel Recommendations”for further information.
- Avoid unnecessary idling.

Shut off the engine rather than idle for long periods of time.

- Observe the service indicator frequently. Keep the air cleaner elements clean.

- Ensure that the turbocharger is operating correctly. For more information refer to this Operation and Maintenance Manual, “Turbocharger - Inspect”.
- Maintain a good electrical system.

One faulty battery cell will overwork the alternator. This fault will consume excess power and excess fuel.

- The belt should be in good condition. Refer to the Systems Operation, Testing and Adjusting, “V - Belt Test” for further information.
- Ensure that all of the connections of the hoses are tight. The connections should not leak.
- Ensure that the driven equipment is in good working order.
- Cold engines consume excess fuel. Utilize heat from the jacket water system and the exhaust system, when possible. Keep cooling system components clean and keep cooling system components in good repair. Never operate the engine without water temperature regulators. All of these items will help maintain operating temperatures.